



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

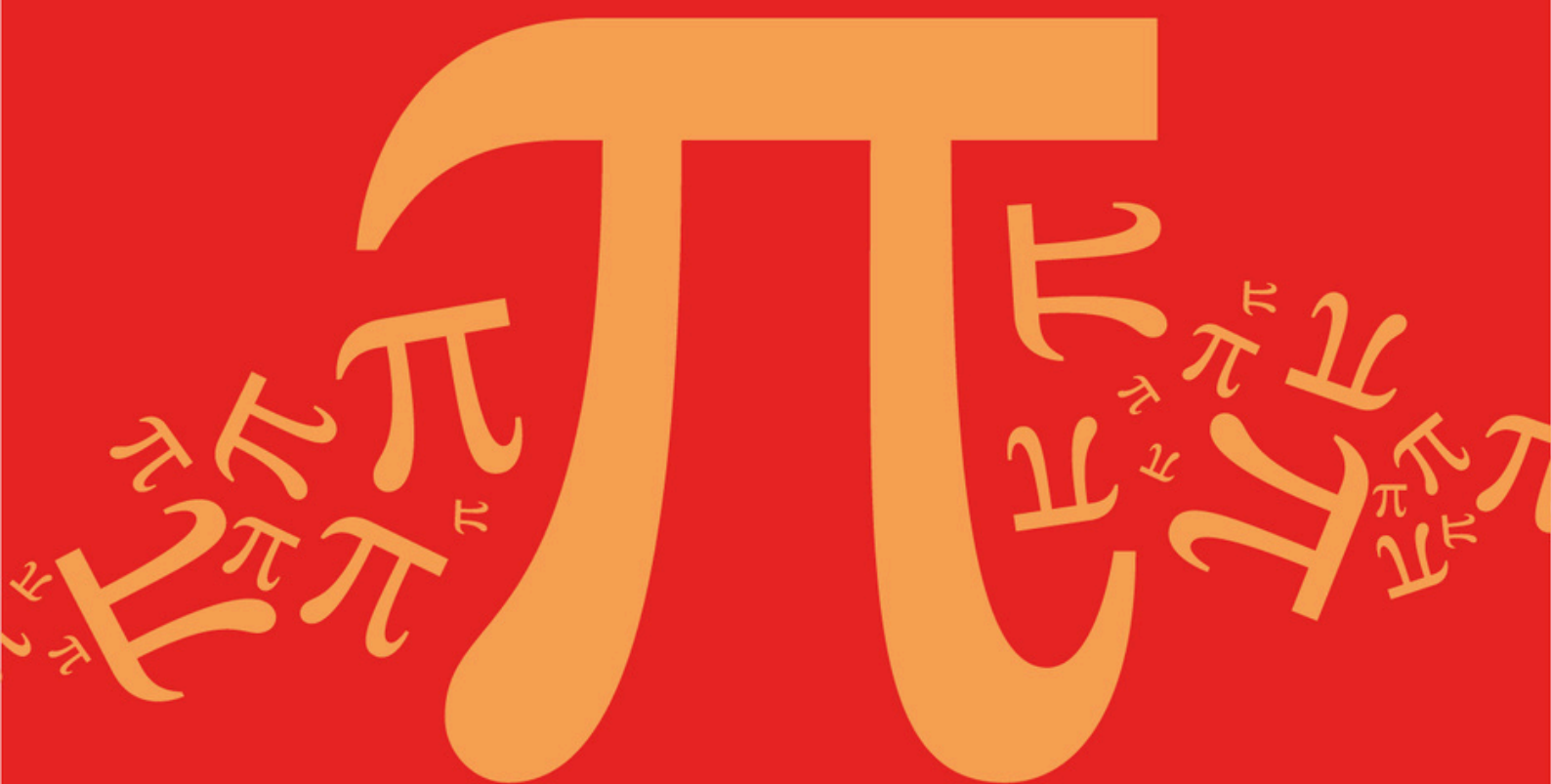
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D7: Evaluate Limits Using Maclaurin

Mark Scheme

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D: Further Algebra and Functions

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D7: Evaluate Limits Using Maclaurin

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

(a) $\sin 2x = 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!}$
Accept ACF even if unsimplified

$$= 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$$

B1

1

(b) $\lim_{x \rightarrow 0} \left[\frac{2x - \sin 2x}{x^2 \ln(1+kx)} \right] = \lim_{x \rightarrow 0} \frac{2x - (2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 \dots)}{x^2 \left(kx - \frac{(kx)^2}{2} + \dots \right)}$
Using series expansions.

M1

Expansion of $\ln(1+kx) = kx (-\dots)$

B1

$$= \lim_{x \rightarrow 0} \left[\frac{\frac{4}{3}x^3 - \frac{4}{15}x^5 + \dots}{kx^3 - \frac{k^2}{2}x^4} \right] = \lim_{x \rightarrow 0} \left[\frac{\frac{4}{3} - O(x^2)}{k - O(x)} \right]$$

Dividing numerator and 0 denominator by x^3 to get constant term in each. Must be at least a total of 3 terms divided by x^3

m1

$$\frac{4}{3k} = 16 \Rightarrow k = \frac{1}{12}$$

OE exact value. Dep on numerator being of form $4/3(OE) + \lambda x^2 \dots (\lambda \neq 0)$ and denominator being of form $k + \mu x \dots (\mu \neq 0)$ before limit taken

A1

4

[5]

Q2.

(a) (i) $\cos x + \sin x = 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3$

Accept coeffs unsimplified, even 3! for 6.

B1
1

(ii) $\ln(1+3x) = 3x - \frac{1}{2}(3x)^2 + \frac{1}{3}(3x)^3 = 3x - \frac{9}{2}x^2 + 9x^3$

Accept coeffs unsimplified

B1
1

(b) (i) $y = e^{\tan x}, \quad \frac{dy}{dx} = \sec^2 x e^{\tan x}$

Chain rule

ACF eg $y \sec^2 x$

$$\frac{d^2 y}{dx^2} = 2 \sec^2 x \tan x e^{\tan x} + \sec^4 x e^{\tan x}$$

Product rule OE

ACF

M1

A1

m1

A1

$$= \sec^2 x e^{\tan x} (2 \tan x + \sec^2 x)$$

$$= \frac{dy}{dx} (2 \tan x + 1 + \tan^2 x)$$

$$\frac{d^2 y}{dx^2} = (1 + \tan x)^2 \frac{dy}{dx}$$

AG Completion; CSO any valid method.

A1
5

(ii) $\frac{d^3 y}{dx^3} = 2(1 + \tan x) \sec^2 x \frac{dy}{dx} + (1 + \tan x)^2 \frac{d^2 y}{dx^2}$

M1

When $x = 0$, $\frac{d^3 y}{dx^3} = 2(1)(1)(1) + (1)(1) = 3$
CSO

A1
2

(iii) $y(0) = 1; y'(0) = 1; y''(0) = 1; y'''(0) = 3;$

$$y(x) \approx y(0) + xy'(0) + \frac{1}{2}x^2 y''(0) + \frac{1}{3!}x^3 y'''(0)$$

M1

$$e^{\tan x} \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3$$

CSO AG

A1

2

(c) $\lim_{x \rightarrow 0} \left[\frac{e^{\tan x} - (\cos x + \sin x)}{x \ln(1 + 3x)} \right]$

$$= \lim_{x \rightarrow 0} \frac{1 + x + \frac{x^2}{2} + \frac{x^3}{2} - 1 - x + \frac{x^2}{2} + \frac{x^3}{6}}{x \left(3x - \frac{9}{2}x^2 + \dots \right)}$$

Using series expns.

M1

$$= \lim_{x \rightarrow 0} \left[\frac{x^2 + \frac{2}{3}x^3 + \dots}{3x^2 - \frac{9}{2}x^3 + \dots} \right] = \lim_{x \rightarrow 0} \left[\frac{1 + \frac{2}{3}x + \dots}{3 - \frac{9}{2}x + \dots} \right]$$

Dividing numerator and denominator by x^2 to get constant terms. OE following a slip.

m1

$$= \frac{1}{3}$$

A1

3

[14]

Q3.

(a) $\frac{dy}{dx} = \frac{2 \sec^2 x}{1 + 2 \tan x}$

Chain rule

M1

ACF for $y'(x)$

A1

$$\frac{d^2 y}{dx^2} = \frac{(1+2\tan x)(4\sec^2 x \tan x) - 2\sec^2 x(2\sec^2 x)}{(1+2\tan x)^2}$$

Quotient rule OE in which both u and v are not const. or applied to a correct form of y'
ACF for $y''(x)$

M1

A1

4

(b) McC. Thm: $y(0) + x y'(0) + \frac{x^2}{2} y''(0)$

$(y(0) = 0); \quad y'(0) = 2; \quad y''(0) = -4$

Attempt to evaluate at least $y'(0)$ and $y''(0)$. PI

M1

$\ln(1+2\tan x) \approx 2x - 2x^2$

Dep on previous 5 marks

A1

2

(c) $\ln(1-x) = -x - \frac{1}{2}x^2 \dots$

Ignore higher power terms

B1

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$$\left[\frac{\ln(1+2\tan x)}{\ln(1-x)} \right] \approx \frac{2x - 2x^2 \dots}{-x - \frac{1}{2}x^2 \dots}$$

Expansions used

M1

$$= \frac{2 - 2x \dots}{-1 - \frac{1}{2}x \dots}$$

Dividing num. and den. by x to get constant term in each and non-const term in at least num. or den.

m1

So $\lim_{x \rightarrow 0} \left[\frac{\ln(1+2\tan x)}{\ln(1-x)} \right] = \frac{2}{-1} = -2$

ft c's answer to (b) provided answer (b) is in the form $\pm px \pm qx^2$ and B1 awarded

A1F

4

[10]

Q4.

(a) $\frac{dy}{dx} = \frac{1}{4+3x} \times 3$
Chain rule

M1

$$\frac{d^2y}{dx^2} = -3(4+3x)^{-2} \times 3 = -9(4+3x)^{-2}$$

M1 for quotient (PI) or chain rule used

M1A1

3

(b) $\ln(4+3x) = \ln 4 + y'(0)x + y''(0)\frac{1}{2}x^2 + \dots$
Clear attempt to use Maclaurin's theorem
with numerical values for $y'(0)$ and $y''(0)$

M1

First three terms: $\ln\left(4 + \frac{3}{4}x - \frac{9}{32}x^2\right)$
ft on c's answers to (a) provided $y'(0)$ and
 $y''(0)$ are $\neq 0$. Accept 1.38(6..) for $\ln 4$

A1F

2

(c) $\ln(4-3x) = \ln 4 - \frac{3}{4}x - \frac{9}{32}x^2$
ft $x \rightarrow -x$ in c's answer to (b)

B1F

1

(d) $\ln\left(\frac{4+3x}{4-3x}\right) = \ln(4+3x) - \ln(4-3x)$

M1

$$\approx \ln 4 + \frac{3}{4}x - \frac{9}{32}x^2 - \ln 4 + \frac{3}{4}x + \frac{9}{32}x^2$$

$$\approx \frac{3}{2}x$$

CSO AG

A1

2

[8]

Q5.

(a) $\sin 3x = 3x - \frac{1}{3!} (3x)^3 + \dots = 3x - 4.5x^3 + \dots$

B1

1

(b) $\cos 2x = 1 - \frac{1}{2!} (2x)^2 + \dots$

B1

$$\lim_{x \rightarrow 0} \left[\frac{3x \cos 2x - \sin 3x}{5x^3} \right] = \lim_{x \rightarrow 0} \frac{3x - 6x^3 - 3x + 4.5x^3 + \dots}{5x^3}$$

Using expansions

M1

$$= \lim_{x \rightarrow 0} \frac{-1.5 + (0(x^2)) \dots}{5}$$

Division by x^3 stage to reach relevant form of quotient before taking limit.

m1

$$= -\frac{3}{10}$$

CSO OE

A1

4

[5]

Q6.

(a) $\cos 4x \approx 1 - \frac{(4x)^2}{2} + \frac{(4x)^4}{4!} \dots$

Clear attempt to replace x by $4x$ in expansion of $\cos x$...condone missing brackets for the M mark

M1

$$\approx 1 - 8x^2 + \frac{32}{3}x^4 \dots$$

A1

2

(b) (i) $\frac{dy}{dx} = \frac{1}{2 - e^x} \times (-e^x)$
Chain rule

M1A1

$$\frac{d^2y}{dx^2} = \frac{(2 - e^x)(-e^x) - (-e^x)(-e^x)}{(2 - e^x)^2}$$

Quotient rule OE

M1

ACF

A1

$$= \frac{-2e^x}{(2 - e^x)^2}$$

$$\frac{d^3y}{dx^3} = \frac{(2 - e^x)^2(-2e^x) - (-2e^x)2(2 - e^x)(-e^x)}{(2 - e^x)^4}$$

*All necessary rules attempted
 (dep on previous 2 M marks)*

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m1

ACF

A1

6

(ii) $y(0) = 0; y'(0) = -1; y''(0) = -2; y'''(0) = -6$
At least three attempted

M1

$$\ln(2 - e^x) \approx y(0) + xy'(0) + \frac{x^2}{2}y''(0) \dots$$

$$\dots \approx -x - x^2 - x^3 \dots$$

*CSO AG (The previous 7 marks
 must have been awarded and no
 double errors seen)*

A1

2

(c)
$$\left[\frac{x \ln(2 - e^x)}{1 - \cos 4x} \right] \approx \frac{-x^2 - x^3 - x^4}{8x^2 - \frac{32}{3}x^4}$$

Using the expansions

M1

$$\text{Limit} = \lim_{x \rightarrow 0} \frac{-x^2 - o(x^3)}{8x^2 - o(x^4)}$$

The notation $o(x^n)$ can be replaced by a term of the form kx^n

$$\dots = \lim_{x \rightarrow 0} \frac{-1 - o(x)}{8 - o(x^2)}$$

Division by x^2 stage before taking the limit

m1

$$\dots = -\frac{1}{8}$$

CSO

A1

3

[13]

Q7.

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(a) (i)
$$e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots$$

Clear use of $x \rightarrow 2x$ in expansion of e^x

M1

ACF

A1

2

(ii)
$$\{f(x)\} = e^{2x}(1 + 3x)^{-\frac{2}{3}}$$

$$(1 + 3x)^{-\frac{2}{3}} = 1 + \left(-\frac{2}{3}\right)(3x) + \frac{\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)(3x)^2}{2} - \frac{40}{3}x^3$$

First three terms as

$$1 + \left(-\frac{2}{3}\right)(3x) + kx^2 \quad \text{OE}$$

M1

$$= 1 - 2x + 5x^2 - \frac{40}{3}x^3$$

A1

$$\{f(x) \approx\}$$

$$1 + 2x + 2x^2 + \frac{4x^3}{3} - 2x - 4x^2 - 4x^3 + 5x^2 + 10x^3 - \frac{40x^3}{3}$$

Dep on both prev MS

m1

Condone one sign or numerical slip in mult.

A1Ft

$$= 1 + 3x^2 - 6x^3$$

CSO AG A0 if binominal series not used

A1

5

(b) (i) $y = \ln(1 + 2 \sin x) \Rightarrow \frac{dy}{dx} = \frac{1}{1 + 2 \sin x} - 2 \cos x$

Chain rule

M1A1

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$$\frac{d^2y}{dx^2} = \frac{(1 + 2 \sin x)(-2 \sin x) - 2 \cos x(2 \cos x)}{(1 + 2 \sin x)^2} = \frac{-2(\sin x + 2)}{(1 + 2 \sin x)^2}$$

Quotient rule OE with u and v non constant

M1

ACF

A1

4

(ii) $y(0) = 0, \quad y'(0) = 2, \quad y''(0) = -4$

M1

McL Thm: $\{\ln(1 + 2 \sin x)\} \approx 0 + 2x - 4 \left(\frac{x^2}{2} \right) + \dots \approx 2x - 2x^2$
 CSO AG

A1

2

(c) $\lim_{x \rightarrow 0} \frac{1 - f(x)}{x \ln(1 + 2 \sin x)} = \lim_{x \rightarrow 0} \frac{-3x^2 + 6x^3}{2x^2 - 2x^3}$

Using expansions

M1

$$= \lim_{x \rightarrow 0} \frac{-3 + 6x}{2 - 2x}$$

*Division by x^2 stage
before taking limit.*

m1

$$= -\frac{3}{2}$$

CSO

A1

3

[16]

Q8.

(a) (i) $f(x) = (9 + \tan x)^{\frac{1}{2}}$
 so $f'(x) = \frac{1}{2}(9 + \tan x)^{-\frac{1}{2}} \sec^2 x$

Chain rule

M1A1

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$$f''(x) = -\frac{1}{4}(9 + \tan x)^{-\frac{3}{2}} \sec^4 x + \frac{1}{2}(9 + \tan x)^{-\frac{1}{2}} (2 \sec^2 x \tan x)$$

Product rule, OE

M1

ACF

A1

4

(ii) $f(0) = 3$

B1

$$f'(0) = \frac{1}{2}(9)^{-\frac{1}{2}} = \frac{1}{6}$$

*Both attempted and at least one correct ft
on c's $f'(x)$ and $f''(x)$*

M1

$$f'(0) = -\frac{1}{4}(9)^{-\frac{3}{2}} = -\frac{1}{108}$$

$$f(x) \approx f(0) + x f'(0) + \frac{1}{2} x^2 f''(0)$$

$$(9 + \tan x)^{\frac{1}{2}} \approx 3 + \frac{x}{6} - \frac{x^2}{216}$$

CSO AG

A1

3

(b)

$$\frac{f(x) - 3}{\sin 3x} \approx \frac{\frac{x}{6} - \frac{x^2}{216} \dots}{3x - \frac{(3x)^3}{3!} \dots}$$

Using series expns.

M1

$$\approx \frac{\frac{1}{6} - \frac{x}{216} \dots}{3 - \dots}$$

Dividing numerator and denominator by x
to get constant term in each

m1

$$\lim_{x \rightarrow 0} \left[\frac{f(x) - 3}{\sin 3x} \right] = \frac{1}{18}$$

A1

3

[10]

Q9.

(a) (i)

$$\ln(1 + 2x) = 2x - 2x^2 + \frac{8}{3}x^3 \dots$$

Use of expansion of $\ln(1 + x)$

M1

Simplified 'numerators'.

A1

2

(ii)

$$-\frac{1}{2} < x \leq \frac{1}{2}$$

B1

1

(b) (i) $y = \ln \cos x \Rightarrow y'(x) = \frac{1}{\cos x} (-\sin x)$

M1

$$y''(x) = -\sec^2 x$$

ACF

A1

$$y''(x) = -2 \sec x (\sec x \tan x)$$

Chain rule OE

M1

$$\{y'''(x) = -2 \tan x (\sec^2 x)\}$$

Ft a slip...accept unsimplified

A1ft

4

(ii) $y'''(x) = -2[\sec^2 x (\sec^2 x) + \tan x (2 \sec x (\sec x \tan x))]$

Product rule OE

ACF

M1

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A1

$$y'''(0) = -2[(1)^2 + 0] = -2$$

Ft a slip

A1ft

3

(iii) $\ln \cos x \approx 0 + 0 + \frac{x^2}{2}(-1) + 0 \frac{x^4}{4!}(-2)$

M1

$$\approx -\frac{x^2}{2} - \frac{x^4}{12}$$

CSO throughout part (b). AG

A1

2

(c) Limit = $\lim_{x \rightarrow 0} \left[\frac{x \ln(1+2x)}{x^2 - \ln \cos x} \right]$

$$= \lim_{x \rightarrow 0} \left[\frac{x(2x - 2x^2 + \dots)}{x^2 - \left(-\frac{x^2}{2} - \frac{x^4}{12} \dots\right)} \right]$$

Using earlier expansions

M1

$$\text{Limit} = \lim_{x \rightarrow 0} \frac{2x^2 - o(x^3)}{1.5x^2 + o(x^4)}$$

The notation $o(x^n)$ can be replaced by a term of the form kx^n

$$= \lim_{x \rightarrow 0} \frac{2 - o(x)}{1.5 + o(x^2)} = \frac{4}{3}$$

Need to see stage, division by x^2

M1A1

3

[15]

Q10.

(a) $\sin 2x \approx 2x - \frac{(2x)^3}{3!} + \dots = 2x - \frac{4}{3}x^3 + \dots$

B1

1

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(b) (i) $\frac{dy}{dx} = \frac{1}{2}(3 + e^x)^{-\frac{1}{2}}(e^x)$

Chain rule

M1A1

$$\frac{d^2y}{dx^2} = \frac{1}{2}e^x(3 + e^x)^{-\frac{1}{2}} - \frac{1}{4}(3 + e^x)^{-\frac{3}{2}}(e^{2x})$$

Product rule OE

M1

OE

A1

$$y'(0) = \frac{1}{4}; \quad y''(0) = \frac{1}{4} - \frac{1}{32} = \frac{7}{32}$$

CSO

A1

$$(ii) \quad y(0) = 2; y'(0) = \frac{1}{4}; y''(0) = \frac{1}{4} - \frac{1}{32} = \frac{7}{32}$$

$$\text{McC. Thm: } y(0) + xy'(0) + \frac{x^2}{2} y''(0)$$

$$\sqrt{3+e^x} \approx 2 + \frac{1}{4}x + \frac{7}{64}x^2$$

CSO; AG

M1A1

2

$$(c) \quad \left[\frac{\sqrt{3+e^x} - 2}{\sin 2x} \right] = \left[\frac{2 + \frac{1}{4}x + \frac{7}{64}x^2 - 2}{2x - \frac{4}{3}x^3} \right]$$

$$= \left[\frac{\frac{1}{4} + \frac{7}{64}x + \dots}{2 - \frac{4}{3}x^2 + \dots} \right]$$

*Dividing numerator and denominator by x
to get constant term in each*

M1

m1

$$\lim_{x \rightarrow 0} \left[\frac{\sqrt{3+e^x} - 2}{\sin 2x} \right] = \frac{\frac{1}{4}}{2} = \frac{1}{8}$$

*Ft on cand's answer to (a) provided of the
form $ax + bx^3$*

A1F

3

[11]

Q11.

$$(a) \quad (i) \quad f'(x) = \frac{1}{2}(1+2x)^{-\frac{1}{2}}(2) = (1+2x)^{-\frac{1}{2}}$$

M1A1

$$f''(x) = -(1+2x)^{-\frac{3}{2}}$$

ft a slip

A1F

$$f'''(x) = 3(1+2x)^{-\frac{5}{2}}$$

A1

4

(ii) $f(x) = (1+2x)^{\frac{1}{2}} \Rightarrow f(0) = 1;$

B1

$f'(0) = 1; f''(0) = -1; f'''(0) = 3$
All three attempted

M1

ft on $k(1+2x)'''$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2}f''(0) + \frac{x^3}{6}f'''(0)$$

$$\dots \approx 1 + x - \frac{x^2}{2} + \frac{x^3}{2}$$

CSO AG

A1F

A1

4

(b) $e^x (1+2x)^{\frac{1}{2}} \approx$

$$\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right) \left(1 + x - \frac{x^2}{2} + \frac{x^3}{2}\right)$$

Attempt to expand needed

M1

$$\approx 1 + x(1+1) + x^2(-0.5+1+0.5)$$

A1

$$+ x^3 \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{2} + \frac{1}{6} \right)$$

$$\approx 1 + 2x + x^2 + \frac{2}{3}x^3$$

CSO

A1

3

(c)
$$e^{2x} = 1 + 2x + \frac{(2x)^2}{2} + \frac{(2x)^3}{6} + \dots$$

$$= 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots$$

B1

1

(d)
$$1 - \cos x = \frac{1}{2}x^2 + \{o(x^4)\}$$

B1

$$\frac{e^x (1+2x)^{\frac{1}{2}} - e^{2x}}{1 - \cos x} = \frac{1 + 2x + x^2 + \frac{2}{3}x^3 - \left[1 + 2x + 2x^2 + \frac{4}{3}x^3\right]}{\frac{1}{2}x^2 + \{o(x^4)\}}$$

Series used

M1

$$\lim_{x \rightarrow 0} \frac{-x^2 + \{o(x^3)\}}{\frac{1}{2}x^2 + \{o(x^4)\}} = \lim_{x \rightarrow 0} \frac{-1 + o(x)}{\frac{1}{2} + o(x^2)} = -2$$

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A1F

$$\lim_{x \rightarrow 0} \frac{-1 + o(x)}{\frac{1}{2} + o(x^2)} = -2$$

ft a slip but must see the intermediate stage

A1F

4

[16]

Q12.

(a) (i) $f(x) = \ln(1 + e^x)$:

$$f(0) = \ln 2$$

B1

$$f'(x) = \frac{e^x}{1 + e^x} \quad f'(0) = \frac{1}{2}$$

Chain rule

M1
A1

$$f''(x) = \frac{(1+e^x)e^x - e^x e^x}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2}$$

Quotient rule OE

M1
A1

$$f''(0) = \frac{1}{4}$$

so first three terms are:

$$f(x) = \ln 2 + \frac{1}{2}x + \frac{1}{4} \frac{x^2}{2!} = \ln 2 + \frac{1}{2}x + \frac{1}{8}x^2$$

CSO; AG

A1

6

$$(ii) \quad f'''(x) = \frac{(1+e^x)^2 e^x - e^x [2(1+e^x)e^x]}{(1+e^x)^4}$$

Chain rule with quotient/product rule

ft on $f''(x) = ke^x (1+e^x)^n$ (integer $n < 0$)

M1
A1ft

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$$f'''(0) = \frac{4-4}{2^4} = 0$$

CSO; AG;

A1

{so coefficient of x^3 is zero}

All previous differentiation correct

3

SC for those not using Maclaurin's theorem: **maximum** of 4/9

$$(b) \quad \frac{1}{2}x + \frac{1}{8}x^2$$

B1

1

$$(c) \quad \ln\left(1 - \frac{x}{2}\right) = \left(-\frac{x}{2}\right) - \frac{1}{2}\left(-\frac{x}{2}\right)^2 + \frac{1}{3}\left(-\frac{x}{2}\right)^3 - \dots$$

B1

1

(d) $\ln\left(\frac{1+e^x}{2}\right) + \ln\left(1-\frac{x}{2}\right) = -\frac{x^3}{24} + \dots$

Uses previous expansions to obtain first non-zero term of the form kx^3

M1

$$x - \sin x \approx x - \left[x - \frac{x^3}{3!} + \dots \right] \approx \frac{x^3}{3!} + \dots$$

B1

$$\left[\frac{\ln\left(\frac{1+e^x}{2}\right) + \ln\left(1-\frac{x}{2}\right)}{x - \sin x} \right] = \frac{-\frac{1}{24}x^3 + \dots}{\frac{1}{6}x^3 + o(x^5)}$$

M1

$$= \frac{-\frac{1}{24}x^3 + \dots}{x^3 \left[\frac{1}{6} + o(x^2) \right]} = \frac{-\frac{1}{24} + \dots}{\frac{1}{6} + o(x^2)}$$

$$\lim_{x \rightarrow 0} \dots = -\frac{1}{4}$$

CSO

A1

4

[15]

Q13.

(a) $\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 \dots$

B1

1

(b) (i) $f(x) = e^{\sin x} \Rightarrow f(0) = 1$

B1

$$f'(x) = \cos x e^{\sin x} \Rightarrow f'(0) = 1$$

M1A1

$$f'(x) = -\sin x e^{\sin x} + \cos^2 x e^{\sin x}$$

Product rule used

$$f'(0) = 1$$

M1A1

$$\text{Maclaurin } f(x) = f(0) + xf'(0) + \frac{x^2}{2} f''(0)$$

$$\text{so 1st three terms are } 1 + x + \frac{1}{2} x^2$$

CSO AG

A1

6

$$(ii) \quad f'''(x) = \cos x (\cos^2 x - \sin x) e^{\sin x} + \{2\cos x (-\sin x) - \cos x\} e^{\sin x}$$

M1A1

$$f'''(0) = 0 \text{ so the coefficient of } x^3 \text{ in the series is zero}$$

CSO AG

SC for (b): Use of series expansions.....max of 4/9

A1

3

$$(c) \quad \sin x \approx x.$$

Ignore higher power terms in sinx expansion

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B1

$$\frac{e^{\sin x} - 1 + \ln(1-x)}{x^2 \sin x} = \frac{-\frac{1}{3} x^3 + o(x^4)}{x^3}$$

Series from (a) & (b) used

M1

Numerator kx^3 (+...)

A1

$$= \frac{-\frac{1}{3} + o(x)}{1 + o(x^2)}$$

Condone if this step is missing

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1 + \ln(1-x)}{x^2 \sin x} = -\frac{1}{3}$$

*On candidate's x^3 coefficient in (a) provided
lower powers cancel*

A1✓

4

[14]

Extra notes:

$$e^{\sin x} = 1 + \left(x - \frac{x^3}{3!} \dots\right) + \frac{1}{2!} \left(x - \frac{x^3}{3!} \dots\right)^2 + \frac{1}{3!} \left(x - \frac{x^3}{3!} \dots\right)^3 \dots$$

M1 for 1st 3 terms ignoring any higher powers than those shown.

A1 for all 4 terms (could be treated separately) ie last term often only comes into (b)(ii)

$$= 1 + x - \frac{x^3}{6} + \frac{1}{2} \left(x^2 - \dots\right) + \frac{1}{6} \left(x^3 - \dots\right) = 1 + x + \frac{1}{2} x^2$$

A1 (be convinced. ignore any powers of x above power 2)

$$\text{Coefficient of } x^3: -\frac{x^3}{6} + \frac{1}{6} x^3 = 0$$

A1 (be convinced....ignore any powers of x above power 3)

Quite often the 2nd A mark is awarded before the 1st A1

Q14.

(a) (i) $(1+y)^{-1} = 1 - y + y^2 \dots$

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B1

1

(ii) $\sec x \approx \frac{1}{1 - \frac{x^2}{2} + \frac{x^4}{24} \dots}$

B1

$$= \left[1 - \frac{x^2}{2} + \frac{x^4}{24} \dots \right]^{-1} =$$

M1

$$\left\{ 1 - \left(-\frac{x^2}{2} + \frac{x^4}{24} \right) + \left(-\frac{x^2}{2} + \frac{x^4}{24} \right)^2 \right\}$$

M1

$$= \left\{ 1 + \frac{x^2}{2} - \frac{x^4}{24} + \frac{x^4}{4} + \dots \right\}$$

$$= 1 + \frac{x^2}{2} + \frac{5x^4}{24}$$

AG be convinced

A1;A1

5

Alternative: Those using Maclaurin

$$f(x) = \sec x$$

$$f(0) = 1; f'(x) = \sec x \tan x; \{f'(0) = 0\}$$

(B1)

$$f''(x) = \sec x \tan^2 x + \sec^3 x; f''(0) = 1$$

Product rule oe

(M1)

$$f'''(x) = \sec x \tan^3 x + 5 \tan x \sec^3 x;$$

Chain rule with product rule OE

(m1)

$$f^{(iv)}(x) = \sec x \tan^4 x + 18 \tan^2 x \sec^3 x \dots$$

$$+ 5 \sec^5 x \Rightarrow f^{(iv)}(0) = 5$$

sec x ≈ printed result

CSO AG

(A2)

(b) $f(x) = \tan x;$

$$f(0) = 0; f'(x) = \sec^2 x; \{f'(0) = 1\}$$

B1

$$f''(x) = 2 \sec x (\sec x \tan x); f''(0) = 0$$

$$f'''(x) = 4 \sec x \tan x (\sec x \tan x) + 2 \sec^4 x$$

Chain rule with product rule oe

M1

$$f'''(0) = 2$$

$$\tan x = 0 + 1x + 0x^2 + \frac{2}{3!}x^3 \dots = x + \frac{1}{3}x^3$$

CSO AG

A1

3

Alternative: Those using otherwise

$$\dots = \frac{\sin x}{\cos x} \approx \left(x - \frac{x^3}{6} \dots \right) \left(1 + \frac{x^2}{2} \dots \right)$$

(M1) (A1)

$$= x + \frac{x^3}{2} - \frac{x^3}{6} \dots = x + \frac{1}{3}x^3 \dots$$

(A1)

(c)
$$\left(\frac{x \tan 2x}{\sec x - 1} \right) = \frac{x(2x + o(x^3))}{\frac{x^2}{2} + o(x^4)}$$

$$\tan 2x = 2x + \frac{1}{3}(2x)^3$$

B1

Condone $o(x^k)$ missing

M1

$$\frac{2 + o(x^2)}{\frac{1}{2} + o(x^2)}$$

M1

$$\lim_{x \rightarrow 0} \left(\frac{x \tan 2x}{\sec x - 1} \right) = 4$$

ft on 2k after B0 for $\tan 2x = kx + \dots$

A1ft

4

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[13]

Examiner reports

Q1.

This question on limits was generally answered well. Most students used the expansion of $\sin x$ from the formulae book and replaced x with $2x$ correctly to find the expansion of $\sin 2x$. In part (b) most students used the correct expansion of $\ln(1 + kx)$. The majority used their expansion from part (a) in the numerator, although some made an algebraic error in the subtraction of $\sin 2x$. Most then correctly divided numerator and denominator by x^3

before taking the limit. Some errors were seen in rearranging the equation $\frac{4}{3k} = 16$ to find the value of k . Not all students showed sufficient terms in the numerator and/or denominator for full marks to be scored.

Q2.

Most candidates were able to write down the required two expansions in parts (a).

In part (b)(i), candidates generally used the chain rule to find $\frac{dy}{dx}$ and then applied the

product rule to find $\frac{d^2y}{dx^2}$. Although a common error was to differentiate $\sec^2 x$ as

$2\sec x \tan x$, those who did find $\frac{d^2y}{dx^2}$ correctly often produced a convincing solution to reach the printed answer.

Some very good solutions were seen for parts (b)(ii) and (b)(iii) but it was disturbing to see some other candidates make no attempt to find the third derivative in part (b)(ii), yet claim that $f'''(0) = 3$ and write down the printed result in part (b)(iii). Clearly in such cases marks cannot be awarded.

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Candidates who attempted part (c) generally showed a thorough understanding of the process required, including the need for explicitly reaching the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$.

Q3.

A majority of candidates were able to differentiate the given function correctly to find the first and second differentials although some made the second differentiation more complicated than was needed by first expressing their answer for the first derivative in terms of $\sin 2x$ and $\cos 2x$. Candidates should be aware that the expression for the derivative of $\sec x$ is in the formulae booklet.

In part (b), most candidates displayed good knowledge of Maclaurin's theorem but only those who had made no errors in earlier differentiations could score both marks for showing the printed result. It is pleasing to report that there was an improvement in candidates explicitly reaching the stage of a constant term with higher order terms in each of the numerator and denominator before taking the limit as $x \rightarrow 0$ in the final part of the question.

Q4.

Most candidates were able to find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ correctly, although some less able candidates failed to apply the chain rule. Although a few candidates in part (b) attempted to use the printed expansion of $\ln(1+x)$ from the formulae booklet instead of applying the prescribed method, the majority of candidates answered the question as instructed and showed good knowledge of Maclaurin's theorem. Many candidates failed to appreciate that in part (c) they had to replace x with $-x$, and instead multiplied both their x and x^2 terms by -1 . A significant minority of candidates in part (d) did not realise that they had to write the expression as the difference of their two expansions.

Q5.

This question on series expansions and the limiting process was generally answered very well, but a significant minority of candidates in part (b) did not explicitly reach the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$.

Q6.

Candidates' differentiation skills have improved significantly over recent series. Almost all candidates could correctly quote the series expansion for $\cos 4x$ although some failed to give their answer in its simplest form. The methods required to obtain the three derivatives were well understood and, although some algebraic errors were seen, there were many correct expressions presented.

Most candidates displayed good knowledge of Maclaurin's theorem but only those who had made no errors in earlier differentiations could score both marks for showing the printed result in part (b)(ii). There was a pleasing improvement in explicitly reaching the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$ in the final part of the question.

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Q7.

Most candidates were able to write down the first four terms in the series expansion of e^{2x} , but in part (a)(ii) a significant minority of candidates ignored the request to use the binomial series expansion and instead resorted to differentiation of $f(x)$ and the use of Maclaurin's theorem. In part (b)(i), the most common error was to fail to use the chain rule

and thus to write the derivative of $\ln(1+2\sin x)$ as $\frac{1}{1+2\sin x}$.

Those candidates who correctly applied the chain rule usually went on to find the correct

expression for $\frac{d^2y}{dx^2}$ by applying either the quotient rule or the product rule. Almost all

candidates who obtained an expression for $\frac{d^2y}{dx^2}$ showed a correct understanding of Maclaurin's theorem in part (b)(ii).

Although the majority of candidates could correctly find the limiting value of the given expression as x tended to 0 in part (c), there were some incorrect solutions seen which just involved substituting $x = 0$ into the given expression.

Q8.

Most candidates were able to find $f'(x)$ correctly although some weaker candidates failed to apply the chain rule and were heavily penalised. The vast majority of the other candidates used the product rule (or quotient rule) but errors in differentiating $\sec^2 x$ were common. The majority of candidates showed good knowledge of Maclaurin's theorem but only those who had made no errors in earlier differentiations could score all 3 marks for showing the printed result in part (a)(i). Many weaker candidates failed to realise that the series expansion for $\sin 3x$ was required in part (b) and just stated the incorrect answer 0 for the limit. A significant minority of other candidates who used the expansion for $\sin 3x$ did not explicitly reach the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$.

Q9.

Although many candidates were able to answer part (a) correctly, it was clear, since factorials were present, that some had not realised that the series expansion for $\ln(1 + x)$ and its range of validity are both given in the formulae booklet. Methods of differentiation, chain rule and product/quotient rules, required in part (b) were either not known or not applied correctly by a significant number of candidates. Some candidates left expressions unsimplified before carrying out further differentiation, often leading to expressions which required multiple applications of the chain, product and quotient rules. Such an approach clearly used up valuable examination time. Those who did simplify their expressions generally scored the available marks for parts (b). Although careless sign errors were seen in solutions to find the limit for part (c), the general understanding of the principles involved was better displayed than in some previous series.

Q10.

The examiners expected candidates to evaluate $3!$ in the expansion for $\sin 2x$, although credit was given retrospectively if the evaluation was left until part (c). Most candidates gave the correct expansion in part (a), and it was pleasing to find a greater proportion of the candidates than last summer applying the chain rule and product rule correctly in part (b)(i). Although Maclaurin's theorem was well known, a significant number of candidates who had obtained a wrong value for the second derivative in part (b)(i) tried to convince the examiners that it led to the printed result in part (b)(ii). Such candidates would have been better advised to look for their error in part (b)(i). Although many candidates scored the three marks in part (c), there were others who did not show the division of the numerator and denominator by x before finding the limit as x tended to zero.

Q11.

The vast majority of candidates answered both parts of part (a) correctly. Those who failed to use Maclaurin's theorem in part (a)(ii) were heavily penalised. Most candidates obtained the printed result in part (b), although others frequently failed to include all the terms in their working. Generally answers to part (c) were correct, although some solutions did not have the correct denominator for the x^3 term. In part (d), weaker candidates just replaced x by 0 and gave the wrong answer " $0/0 = 1$ ". In answering part (d), candidates were expected to show the division of terms by x^2 in the numerator and denominator so that the limiting value can be justifiably stated.

Q12.

Many candidates, including those who had scored almost full marks on the previous five questions, failed to score many marks on this question. In part (a), a minority of candidates attempted to use series expansions instead of the prescribed Maclaurin's

theorem. Although a special case was allowed, with a significant reduction in available marks, very few of these candidates scored any of them. For those using Maclaurin's theorem, most scored the first mark for finding $f(0)$ but a large number of candidates failed

to use the chain rule and differentiated $\ln(1 + e^x)$ incorrectly as $\frac{1}{1+e^x}$. Some of these candidates 'recovered' to score the next method mark but others just continued incorrectly to give subsequent derivatives in the form $(1 + e^x)^n$. Part (b) was not answered well, the

most common error being illustrated by

$$\ln\left(\frac{1+e^x}{2}\right) = \ln\frac{2}{2} + \frac{1}{2}\left(\frac{1}{2}x\right) + \frac{1}{2}\left(\frac{1}{8}x^2\right) = \frac{1}{4}x + \frac{1}{16}x^2$$

Most candidates gave the correct answer for part (c) and in part (d) a significant majority of candidates scored at least half the marks. Some candidates failed to use sufficient terms in the series expansion of $\sin x$.

Q13.

A common error in part (a) was to write the expansion of $\ln(1 - x)$ as the expansion of $-\ln(1 + x)$. It was disappointing to see so many candidates failing to use Maclaurin's theorem to answer part (b) especially as almost an identical question appears in the specimen paper for this unit. Those who did differentiate $f(x)$ generally obtained the first two derivatives correctly but then had difficulty when differentiating $\cos^2 x e^{\sin x}$. A significant minority of candidates failed to use the approximation for $\sin x$ in part (c).

Q14.

Average and weaker candidates scored relatively few marks in this question. The vast majority of candidates gained the mark for their answer to part (a)(i) but in part (ii) a common error for the lower grade candidate was to replace y by $\cos x$ and write ' $\sec x = 1 - \cos x + \cos^2 x \dots$ ' and then insert the expansion for $\cos x$. Those who used Maclaurin's theorem in part (a)(ii) were generally able to find correct expressions for $f'(x)$ and $f''(x)$ but struggled to find $f'''(x)$ and $f^{(iv)}(x)$. Those who attempted part (b) generally gained the first mark for a correct start and first derivative but a significant minority omitted a '2' when finding $f'(x)$ as $\sec^2 x \tan x$. Those candidates who started by writing $\tan x = \sin x \sec x$ and applying the expansions for $\sin x$ and $\sec x$ generally obtained the printed answer convincingly. In part (c) the expansion for $\tan 2x$ was generally correct and most used the relevant expansions but then a significant number of candidates did not show the division of both the numerator and denominator by x^2 .